CARL-Ω — 10K SWEET SPOT BUDGET SURVEY WORKSHEET

FLAGSHIP PHYSICAL MAKER & RESEARCH PROTOTYPE • SUB-₹10,000 INR CEILING

10K SWEET SPOT | BUDGET SURVEY | ZERO-SOLDER

Target Ceiling: ₹10,000 INR | Sim-to-Real: `vessel_kinetic.xml`

10K ARCHITECTURAL GOAL: Dual-processor embodiment (Orange Pi 5 6.0 TOPS NPU for 60FPS YOLOv8 vision + ESP32 PNS for microsecond motor PWM), 6x Solid-State ToF LiDAR array, current sensing & active head gaze. **Market:** Koti, Hyderabad

10K BUDGET SURVEY SCRIPT FOR KOTI COUNTERS (Universal / Sagar / CTC Computer Dealers):

- *"Bhaiya, M.Tech/R&D thesis project ke liye **Orange Pi 5 (RK3588S) ya Raspberry Pi 4** aur accompanying sensors ka **rough quotation estimate** bana rahe hain."*
- *"Agar Orange Pi 5 stock mein nahi hai, toh **Orange Pi 3B (with NPU) ya Pi Zero 2 W** ka kya rate padega?"*
- *"Mujhe soldering nahi aati, saare sensors (VL53L1X, PCA9685, INA219, IMU) **pre-soldered pins ya screw terminals** ke saath chahiye."*

1. High-Level AI Compute & Co-Processor (Brain Subsystem)

Target Range: ₹6,900 – ₹7,300

#	COMPONENT	SPECIFICATION & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK STATUS
1	**Orange Pi 5 (4GB RAM)** Rockchip RK3588S, 6.0 TOPS NPU ✓ STANDALONE SBC	High-level brain: runs local quantized 60FPS YOLOv8 on NPU + CANN spatial torus memory directly on-chassis. 👁 Ask: "Orange Pi 5 4GB RAM ya equivalent 6 TOPS NPU board mil jayega?" *Fallback A: Orange Pi 3B 2GB/4GB (0.8 TOPS NPU, ₹3,800–₹4,200).* *Fallback B: Raspberry Pi 4 4GB (₹4,800) / Pi Zero 2 W (₹1,650).*	**1**	₹6,600–₹6,900			[ ] In Stock [ ] Order (2-day)
2	**ESP32 Dev Board (30-pin)** NodeMCU / WROOM-32 ✓ PRE-SOLDERED PINS	PNS Co-processor: microsecond motor PWM, I2C telemetry polling, UART link to Orange Pi. 👁 Ask: "30-pin narrow ESP32 board with pre-soldered pins rate kya hai?"	**1**	₹300–330			[ ] Soldered
3	**USB-C Power Cable / Supply** 5V 4A / Type-C Power Lead ✓ PLUG & PLAY USB-C	Feeds high-current clean 5V to Orange Pi 5 to prevent brownouts under 6 TOPS NPU AI load.	**1**	₹90–140			[ ] Available

2. Solid-State Micro-LiDAR Array & Vision Perception

Target Range: ₹1,550 – ₹1,750

#	COMPONENT	SPECIFICATION & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK STATUS
4	**VL53L1X Laser ToF** Solid-State Micro-LiDAR (4m) ✓ PRE-SOLDERED PINS	Up to 4.0m range, 940nm Class-1 laser pulses. Millimeter precision in pitch-black rooms. 👁 Ask: "VL53L1X (4 meter range) per-piece rate kitna hai? 6 pieces ka bundle discount?" *Fallback: VL53L0X (2m range, ₹150/ea) saves ~₹180 total on the array.*	**6**	₹1,020–₹1,140 (~₹180/ea)			[ ] Pre-soldered [ ] Loose pins
5	**720p HD USB Web Camera** Wide-Angle Driverless USB ✓ STANDARD USB-A PLUG	Feeds 30–60 FPS live video to Orange Pi NPU for open-vocabulary grounding and Gemini Robotics API. 👁 Ask: "Wide-angle driverless USB camera module (mini PCB or webcam) rate kya hai?"	**1**	₹450–520			[ ] Available
6	**TCA9548A I2C Mux** 8-Channel I2C Bus Multiplexer ✓ PRE-SOLDERED PINS	Channels all 6x VL53L1X laser sensors without I2C address collision.	**1**	₹75–90			[ ] Soldered

🔍 Single-Board Computer (SBC) Survey Tip in Hyderabad

If Orange Pi 5 is not on the shelf in Gujarati Galli: Universal Electronics can arrange it within 24 hours from distributor, or you can walk 15 mins to **CTC (Chenoy Trade Centre, Parklane, Secunderabad)** where SBC distributors have Orange Pi / Raspberry Pi stock on spot.

CARL-Ω — SERVOS, FORCE SENSORS & LOCOMOTION

12-BIT PWM CONTROLLER, PRESSURE SENSORS, CHASSIS & DUAL SUSPENSION

PAGE 2 SURVEY | SERVO & SENSOR SOCKETS

Target Subtotal: ₹1,850 – ₹2,150

3. Jitter-Free Servo Controller, Pan/Tilt Head & Gripper Servos

Target Range: ₹800 – ₹950

#	COMPONENT	SPECIFICATION & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK STATUS
7	**PCA9685 16-Ch PWM** 12-Bit I2C Servo Controller ✓ STANDARD 3-PIN SERVO SOCKETS	4,096 discrete PWM steps over I2C. Eliminates jitter on all neck & arm servos. Zero software timing overhead. 👁 Ask: "PCA9685 16-channel servo driver board hai kya?"	1	₹170–190			[ ] Available
8	**MG996R Neck Servo** Metal Gear High-Torque (11kg·cm) ✓ 3-PIN FEMALE SERVO PLUG	High-torque metal gear servo for active camera/head pan-tilt gaze scanning without gear strip. 👁 Ask: "Original metal gear MG996R servo rate kya hai?"	1	₹280–320			[ ] Available
9	**SG90 Micro Servos (9g)** Bilateral Reach & Claw Actuators ✓ 3-PIN FEMALE SERVO PLUG	Lightweight micro servos for arm joint articulation and tactile claw grasping. 👁 Ask: "SG90 servos (TowerPro) 4 pieces ka packet rate kya hai?"	4	₹300–350 (~₹75-85/ea)			[ ] Available

4. Tactile Force Skin & Motor Load Telemetry (Fatigue Sensing)

Target Range: ₹420 – ₹480

#	COMPONENT	SPECIFICATION & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK STATUS
10	**FSR-402 Pressure Pads** Force Sensitive Resistors (Fingertips) ✓ 2.54MM BREADBOARD PUSH-IN	Polymer tactile skin reading claw grip force (0.1N to 10N) to prevent crushing fragile objects. 👁 Ask: "FSR-402 pressure / force sensor pad available hai kya?"	2	₹300–350 (~₹160/ea)			[ ] Available
11	**INA219 Current Sensor** High-Side I2C Power Monitor ✓ SCREW TERMINAL BLOCK	Measures live motor current (I ∝ Torque) → feeds `carl_fatigue.py` to trigger organic limping. 👁 Ask: "INA219 I2C current sensor with screw terminals rate kya hai?"	1	₹110–130			[ ] Available
12	**MPU6050 IMU Sensor** 6-DOF Accel + Gyroscope ✓ PRE-SOLDERED PINS	Measures yaw angular rate and slip acceleration. Grounds CANN head-direction ring.	1	₹130–145			[ ] Soldered

5. Locomotion Deck, Dual Spring Casters & High-Current Power

Target Range: ₹980 – ₹1,150

#	COMPONENT	SPECIFICATION & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK STATUS
13	**2WD Smart Chassis + Encoders** Base Deck + BO Motors + Discs ✓ BOLT-TOGETHER ASSEMBLY	Acrylic plate, 2x 1:48 BO motors with optical encoder slotted discs, 2x 66mm rubber wheels. 👁 Ask: "2WD chassis with encoder speed discs rate kya hai? Motors wired hain?"	1 kit	₹380–440			[ ] With Encoders
14	**Dual CY-15A Spring Casters** Front & Rear Cushioned Ball Units ✓ SCREW MOUNT	Front & rear suspension compliance ($k=350 ext{ N/m}$). Eliminates teeter-totter hull pitch oscillation. 👁 Ask: "CY-15A metal ball casters 2 pieces rate kya hai?"	2	₹120–160 (~₹75/ea)			[ ] Available
15	**L298N Motor Driver** Dual H-Bridge with Heat Sink ✓ HEAVY SCREW TERMINALS	Heavy-duty red board with aluminum heatsink and screw terminal blocks for motor wires and battery.	1	₹110–135			[ ] Screw Blocks
16	**High-Drain 18650 Cells + 2S Box** 2x 2600mAh High-Discharge ✓ SNAP-IN CELLS	High-drain cells capable of handling simultaneous Orange Pi NPU burst + 5 servos + 2 drive motors.	1 set	₹260–320			Brand: ____
17	**LM2596 5V 3A Buck Regulator** Synchronous Step-Down ✓ SCREW TERMINALS	Delivers solid 5.0V regulated power to Orange Pi 5 and ESP32 with zero voltage sag.	1	₹75–90			[ ] Screw Blocks
18	**Dupont Wires & Breadboard** 3x 40-pin Ribbons + 400-pt Board ✓ PLUG & SOCKET BUS	F-F, M-F, M-M jumper ribbons + mini breadboard + CH-2 spring lever clamps.	1 set	₹150–180			[ ] Complete Set
💰	**GRAND TOTAL ESTIMATED 10K SWEET SPOT EMBODIMENT (18 ITEMS)**			**₹9,650–10,250**	**₹ _____**	**₹ _____**	**≤ ₹10,000 Target**

CARL-Ω — PROCUREMENT STRATEGY & SUB-10K TIERS

TWO-STAGE PHASED BUYING PLAN, KOTI SBC SHOPS & COUNTER CHECKLIST

📍 Where to Inquire for Single-Board Computers (SBC)

- **In Gujarati Galli / Troop Bazaar (Koti):**
  - **Universal Electronics (Gujarati Galli):** Largest selection of development boards; can source Orange Pi 5 or Pi 4 from their central warehouse within 24 hours.
  - **Sagar Electronics (Gujarati Galli):** Competitive pricing on all sensors, PCA9685, INA219, and motors.
  - **Vishal Electronics (Troop Bazaar):** Chassis, dual CY-15A casters, BO gearmotors with encoder discs.
- **Nearby Specialized Computer Markets (If SBC not in Koti):**
  - **Chenoy Trade Centre (CTC), Parklane, Secunderabad:** 15-20 min metro ride from Koti. Hyderabad's central IT hardware hub with direct Pi/Orange Pi distributors.
  - **Online Sourcing (Robu.in / Silverline Electronics):** If Koti quotes >₹7,200 for Orange Pi 5, Robu.in delivers to Hyderabad in 2-3 days at ₹6,800.

💡 The Smart Two-Phase Procurement Strategy

- **Phase 1 (The Sensorimotor Mobility Foundation — Month 1):**
  - Buy the 2WD chassis kit, BO motors, ESP32, 3x ToF sensors, L298N driver, battery pack & breadboard.
  - **Phase 1 Out-of-Pocket: ₹2,150 – ₹2,350 INR.**
  - You validate Stage 1 Body Babbling, Stage 2 CANN, and Stage 3.4 MPPI steering immediately using laptop Wi-Fi telemetry!
- **Phase 2 (The High-Compute NPU & Vision Upgrade — Month 2):**
  - Once the chassis is rolling smoothly, purchase the Orange Pi 5, 720p USB camera, PCA9685, and servos.
  - **Phase 2 Out-of-Pocket: ₹7,500 – ₹7,900 INR.**
  - Mounts directly on top of the tested rolling chassis! Zero risk of buying expensive compute before kinematics work.

3 Formal Embodiment Tiers under the ₹10,000 Budget Window

Dissertation Budget Scenarios

Tier 1: Frugal Edge Organism (Sub-₹4.5K) — Est: ₹4,100 – ₹4,600

Raspberry Pi Zero 2 W (Quad-Core 64-bit @ 1.0 GHz) + ESP32 + 3x ToF + 2WD Chassis + Dual Casters.

- 100% standalone onboard Linux host running Python 3 without external PC.
- Zero tethering required during maze traversal.
- Ultra-low power drain (2-3 hour runtimes on dual 18650 pack).

Tier 2: The Final Sweet Spot Flagship (Sub-₹10K) — Est: ₹9,650 – ₹10,200

Orange Pi 5 (6 TOPS NPU) + 6x VL53L1X Micro-LiDAR + 720p Cam + PCA9685 + Servos + INA219.

- Local 60 FPS YOLOv8 NPU object detection + Google Gemini Robotics API streaming.
- Solid-state laser micro-LiDAR array measuring 360° spatial boundaries in pitch darkness.
- Real-time joint current load telemetry for organic limping dynamics under load.

🔍 Counter Testing & Warranty Questions for 10K Items

- **Orange Pi 5:** Ask: "*Board DOA (Dead-on-Arrival) testing guarantee hai na? Agar power nahi aayi toh return/exchange policy kya hai?*" (Reputable shops provide 7-day testing replacement).
- **PCA9685 & Servos:** Ensure PCA9685 board includes screw power terminal for servo rail and 3-pin headers. Check that servo motor horns and mounting screws are inside the polybag.
- **VL53L1X Laser Sensors:** Inspect protective amber film over the tiny dual optical apertures. Ensure it is not scratched or peeled off.
- **Dual CY-15A Casters:** Roll the ball with your thumb—it must roll freely in all directions without gritty resistance or metal grinding.

📝 Market Survey Quotation Notes (Shop Name & Contact Person):